

STRUCTURAL INTEGRITY RESERVE STUDY (FISCAL YEAR: 2024)

BARWOOD CONDOMINIUM II BOCA RATON, FL



Prepared for:

Barwood Condominium II Association, Inc.

Prepared by:



1815 Purdy Avenue, Miami Beach, FL 33139
(305) 250-2936 | www.criterion-cromer.com

November 29, 2023

This item has been digitally signed and sealed by Casey Cromer, P.E. on the date listed above.

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TABLE OF CONTENTS

Introduction	2
Executive Summary	3
1.0 Study Information	4
1.1 Property Description	4
1.2 Sources of Information	4
1.3 Component Notes.....	4
1.4 Key Terms.....	5
2.0 Funding Methodologies	6
2.1 Component Method (Straight Line Funding)	6
2.2 Cash Flow Method (Pooled Funding)	6
3.0 Structural Integrity Reserve Study – Funding Plans.....	7
3.1 Component Method (Straight Line Funding)	7
3.2 Cash Flow Method (Pooled Funding)	7
4.0 Discussion of Funding Plans	12
4.1 Funding Considerations	12
4.2 Limitations of a Structural Integrity Reserve Study	12
5.0 Scope And Basis	13
5.1 Objectives	13
5.2 Level of Service	14
5.3 Opinions of Useful Life.....	14
5.4 Estimating.....	15
5.5 Information Provided	16
6.0 Annual Projected Expenditures	17
7.0 Standards and Limitations	20
8.0 Conclusion	21
Appendix A – Component Photographs	
Appendix B – Reference Document Excerpts	
Appendix C – Qualifications	

INTRODUCTION

Criterion-Cromer Engineers (Criterion) has conducted a Structural Integrity Reserve Study of the Barwood Condominium II Association, Inc. (the "Association") community located at 23279 Barwood Lane North, Boca Raton, FL 33428. The services performed are consistent with our proposal dated September 13, 2023.

Criterion-Cromer Engineers presents this confidential report for the Association's review and use. This report and all Appendices must be reviewed in its entirety to understand our findings and their limitations.

We have conducted our study in general accordance with the Florida Statutes, where applicable, as well as following the methodologies included within the National Reserve Study Standards published by the Community Association Institute (CAI). Our analysis has been conducted by licensed Professional Engineers and other qualified staff working with the advisement of a Reserve Specialist. Casey Cromer, P.E. performed a visual site inspection on October 3, 2023. Excerpts of relevant referenced standards are available in Appendix B. Qualifications of the project team leaders can be found in Appendix C.

In reviewing the engineering assumptions, cost estimates, and projected reserve balance values herein, please understand that their accuracy diminishes greatly beyond Year 5. Long-term projections are intended only to indicate the likely pattern of reserve expenditures and to guide financial planning. Criterion agrees with CAI's recommendation that reserve studies should be updated regularly to allow periodic adjustment of funding plans and strategies.

This report contains one level of analysis: A Structural Integrity Reserve Study.

- Analysis related to the Structural Integrity Reserve Study is presented in Section 3.0.
- Additional information related to scope and basis for these analyses is presented in Section 5.0.

EXECUTIVE SUMMARY

COMMUNITY INFORMATION

Community Name:	Barwood Condominium II Association, Inc.
Jurisdiction:	Palm Beach County, FL
Next Fiscal Year Begins:	January 1, 2024
Starting Reserve Balance (Projected):	\$69,320
Current Annual Contribution:	\$17,280
Projected Rate of Annual Inflation:	3.5%
Projected Annual Return on Investment:	1.5%
Engineer in Responsible Charge:	Casey Cromer, P.E.
Date on Site:	October 3, 2023

FUNDING STATUS AND FUNDING PLANS

Structural Integrity Reserve Study (see Section 3.0 for complete data)

Component Method (Straight Line Funding)

Fully Funded Balance:	\$325,281
Current Percent Funded:	21%

Cash Flow Method (Pooled Funding), Year 1 Annual Contribution:

Baseline Funding Plan:	\$34,130
Full Funding Plan:	\$58,838

1.0 Study Information

1.1 Property Description

The community consists of a four-story building with 36 residential units. The building was originally completed in 1978 and is located in Palm Beach County, Florida.

We understand that the Association is generally responsible for the infrastructure, building exterior, roofing, common area windows and doors, exterior walkways, waterproofing, railings, interior common areas, elevators, and common mechanical, electrical, plumbing, and fire protection systems.

We visited the property on October 3, 2023. During our site visit, we observed the common areas of the property including the rooftop, exterior corridors, electrical room, mechanical room, and other common components. The common components of the property were found to be in generally good condition and appropriately maintained.

1.2 Sources of Information

The following people provided information which may have been utilized in our study:

- Ian Land (Association management)
- Mike Mihaylov (Association)

The following documents were provided to us and reviewed:

- Reserve Study Questionnaire, completed by Ian Land on 9/29/2023
- Proposed budget for 2023
- Condominium documents, various dates

1.3 Component Notes

General notes, comments, assumptions, and recommendations are listed below. If there are any questions or if any of our assumptions are not accurate, please advise us so we can make appropriate changes.

- The State of Florida requires milestone inspections to be performed periodically. The first milestone inspection is typically required to be performed prior to the completion of the building's 30th year. This process involves a thorough structural evaluation and often results in some form of required structural repairs. Although future structural repairs are not predictable, we have included a reasonable allowance for structural repairs associated with the future milestone inspections.
- Based on our communications and review, we understand the Association has previously applied a silicone coating to the preexisting roofing system. If properly inspected, maintained, and repaired, silicone coatings are often warranted to add an additional 15 years to the useful life of the underlying roofing system. Our analysis considers a complete roofing replacement 15 years after

the application of the existing silicone coating (Year 9). Our analysis also considers the application of a new silicone coating to the roofing system in Year 29.

- Our quantity estimations have been performed with the assistance of historical building drawings which may or may not be accurate. While we make every effort to be as accurate as possible, our cost and quantity estimates are to be considered preliminary and by no means a guarantee. We do not recommend using our estimates for future contractor bidding purposes.
- There are components which are excluded from this analysis as their remaining useful life is estimated to be longer than the study period and/or their useful life cannot be reasonably anticipated. Once these components remaining useful life is less than the study period, or their need for repair becomes evident, they should be added to the funding plan calculations. These components include:
 - Common area railings
 - Plumbing supply piping (generally observed to be copper)
 - Plumbing waste piping (generally observed to be PVC, or similar)

Sample component photos are included in Appendix A – Component Photographs.

1.4 Key Terms

Estimated Quantity: The estimated total quantity for a component, typically designated with one of the following units of measurement:

Allowance = a lump sum cost utilized when future expenditures are difficult to predict

Lot = a lump sum cost utilized for or group of items or related work package

EA = each LF = linear foot SF = square foot SQ = roofing square (100 square feet)

Current Cost: The estimated cost to perform the work, on a unit and/or total basis.

Useful Life (UL): The estimated time, in years, that a newly constructed/installed component can be expected to serve its intended function presuming proper construction, and proactive, planned, preventive maintenance.

Remaining Useful Life (RUL): The estimated time, in years, that an existing component can be expected to service its intended function, presuming timely preventive maintenance. Projects expected to occur in the initial year have a remaining useful life of 0.

2.0 Funding Methodologies

2.1 Component Method (Straight Line Funding)

The Component Method is a reserve funding method which is based on the sum of funding for each individual component. This method utilized the fully funded balance as well as the annual contribution rate required to maintain the fully funded balance. The fully funded balance for each component is the direct proportion of the fraction of life "used up". This value is calculated for each component, and then totaled to determine the fully funded balance for the reserve account. The Component Method is based on the current cost and condition of each component and therefore does not consider inflation, returns on investments, or the available excess cash flow that the reserve account may have at any given point.

2.2 Cash Flow Method (Pooled Funding)

The Cash Flow Method is a reserve funding method which is based on the expected balance of the reserve account over time. This method considers the projected inflation rate as well as future returns on investments. The Cash Flow Method can consider various funding goals including baseline funding, threshold funding, and full funding. Cash Flow Method funding plans are typically created based on the following funding goals, as defined by the National Reserve Study Standards (listed by most aggressive to most conservative):

Baseline Funding

Establishing a reserve funding goal of allowing the reserve cash balance to approach but never fall below zero during the cash flow projection. This is the funding goal with the greatest risk of being prepared to fund future repair and replacement of major components, and it is not recommended as a long-term solution/plan. Baseline funding may lead to project delays, the need for a special assessment, and/or a line of credit for the community to fund needed repairs and replacement of major components.

Threshold Funding

Establishing a reserve funding goal of keeping the reserve balance above a specified dollar or percent funded amount. Depending on the threshold selected, this funding goal may be weaker or stronger than "fully funded" with respective higher risk or less risk of cash problems. In determining the threshold, many variables should be considered, including things such as investment risk tolerance, community age, building type, components that are not readily inspected, and components with a remaining useful life of more than 30 years.

Full Funding

Setting a reserve funding goal to attain and maintain reserves at or near 100 percent funded. Fully funded is when the actual or projected reserve balance is equal to the fully funded balance.

3.0 Structural Integrity Reserve Study – Funding Plans

A Structural Integrity Reserve Study only covers a portion of the Association components. Therefore, if one of Structural Integrity Reserve Study funding plans is followed, it will only be predictive if the future expenditures are limited to those presented in the component list for the Structural Integrity Reserve Study. See Section 4.0 for additional commentary.

For this Association the following funding methods/plans have been calculated, as summarized below:

3.1 Component Method (Straight Line Funding)

The Component Method has been calculated for the items included within the scope of a Structural Integrity Reserve Study.

3.2 Cash Flow Method (Pooled Funding)

Cash Flow Method funding plans have been calculated for the items included within the scope of a Structural Integrity Reserve Study. Funding plans have been calculated for the following funding goals:

Baseline Funding Plan

In this plan, the reserve balance remains positive (above zero) throughout the study period. This is the most aggressive funding plan and leaves no margin for changes in costs, inflation, condition, future discrepancies, etc. and results in a likely chance of unplanned special assessments in the future.

Full Funding Plan

In this plan, the reserve balance reaches the projected fully funded balance by the end of Year 7, and then continues to maintain a balance above the projected fully funded balance throughout the study period. This funding plan is the most conservative and results in the least likely chance of unplanned special assessments in the future.

COMPONENT METHOD

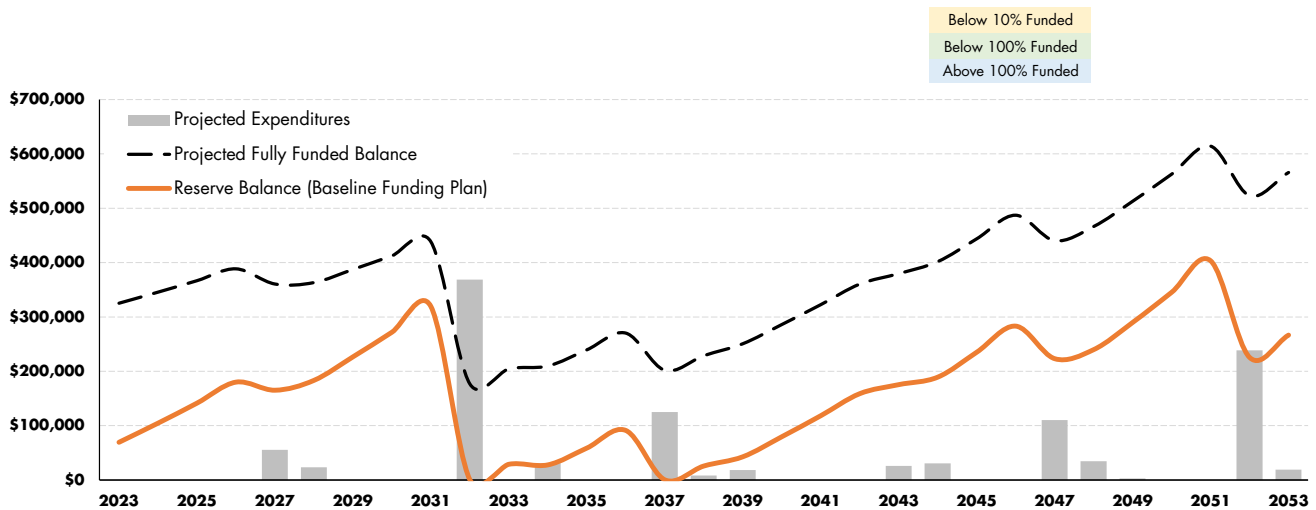
STRUCTURAL INTEGRITY RESERVE STUDY

#	Component Description	Estimated Quantity	Current Cost		UL	RUL	Fully Funded	
			Unit	Total			Balance	Annual Contribution
Building Structure and Envelope								
101	Roofing and appurtenances, replace	144 SQ	\$ 1,900	\$ 273,600	35	8	\$ 211,063	\$ 7,817
102	Roofing, silicone coating	144 SQ	\$ 550	\$ 79,200	35	28	\$ 15,840	\$ 2,263
103	Waterproof coating, walkways	2,650 SF	\$ 4.50	\$ 11,925	15	13	\$ 1,590	\$ 795
104	Exterior painting, waterproofing, and minor repairs	1 Allow	\$ 50,000	\$ 50,000	10	3	\$ 35,000	\$ 5,000
105	Exterior windows, storage rooms	1 Lot	\$ 15,100	\$ 15,100	35	24	\$ 4,746	\$ 431
106	Exterior metal doors, mechanical rooms	4 EA	\$ 950	\$ 3,800	40	20	\$ 1,900	\$ 95
107	Structural repairs, milestone inspection	1 Allow	\$ 18,000	\$ 18,000	20	13	\$ 6,300	\$ 900
Building Structure and Envelope Subtotal (7 Line Items)							\$ 276,439	\$ 17,301
Electrical Systems								
201	Main circuit breaker, 1200 amps, 120/208V, 3P, 4W	1 EA	\$ 11,500	\$ 11,500	40	20	\$ 5,750	\$ 288
202	Electrical subpanels, 60-225 amps	2 EA	\$ 3,000	\$ 6,000	40	15	\$ 3,750	\$ 150
203	Electrical disconnects, electrical room, 60-225 amps	4 EA	\$ 1,250	\$ 5,000	40	15	\$ 3,125	\$ 125
204	Electrical disconnects, elevator room, 60-225 amps	1 EA	\$ 1,250	\$ 1,250	40	25	\$ 469	\$ 31
Electrical Systems Subtotal (2 Line Items)							\$ 13,094	\$ 594
Plumbing								
301	Commercial water heater, 18 kW	1 EA	\$ 10,500	\$ 10,500	20	10	\$ 5,250	\$ 525
302	Vertical storage tank, 119 gallons	1 EA	\$ 5,200	\$ 5,200	20	14	\$ 1,560	\$ 260
303	Domestic water, backflow preventer	1 EA	\$ 3,500	\$ 3,500	40	10	\$ 2,625	\$ 88
304	Domestic water, backflow preventer	1 EA	\$ 6,500	\$ 6,500	40	8	\$ 5,200	\$ 163
Plumbing Subtotal (4 Line Items)							\$ 14,635	\$ 1,035
Fire Protection Systems								
401	Fire alarm panel	1 EA	\$ 13,500	\$ 13,500	15	4	\$ 9,900	\$ 900
402	Fire alarm system components	1 Allow	\$ 7,000	\$ 7,000	25	4	\$ 5,880	\$ 280
403	Fire department connections	1 Allow	\$ 8,000	\$ 8,000	30	10	\$ 5,333	\$ 267
Fire Protection Systems Subtotal (3 Line Items)							\$ 21,113	\$ 1,447
Total / Fully Funded Balance							\$ 325,281	\$ 20,377

BASELINE FUNDING PLAN

STRUCTURAL INTEGRITY RESERVE STUDY

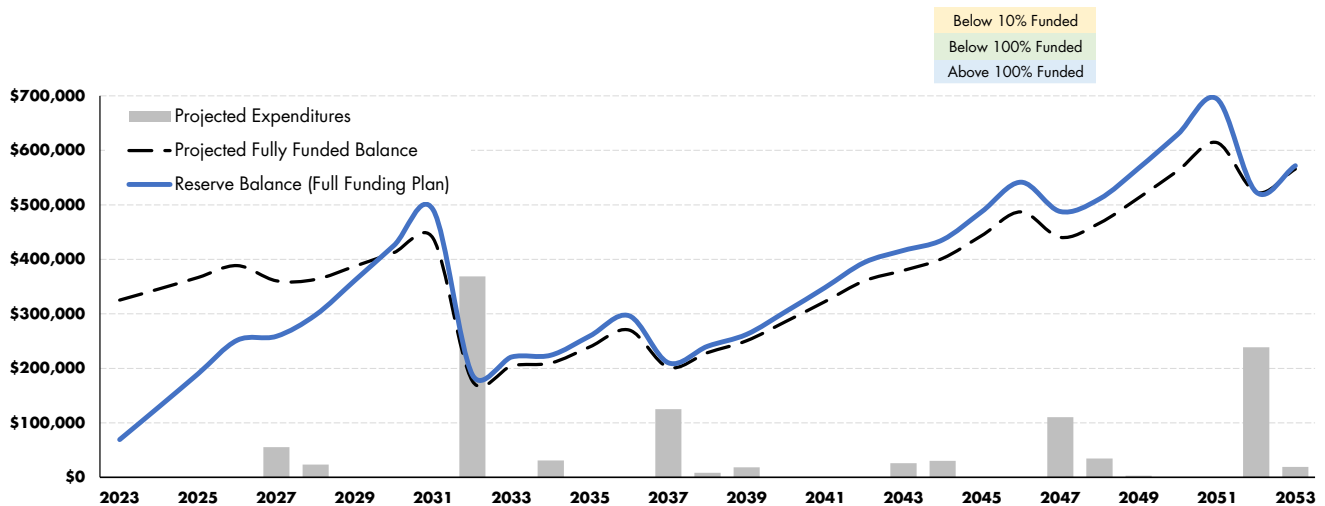
Year		Reserve Balance (Start of Year)	Investment Returns	Projected Expenditures	Reserve Contributions			Reserve Balance (End of Year)	Component Method	
					Association Annual Total	Average Unit Monthly Total	Change from Prior Year		Fully Funded Balance	Percent Funded
2023	Year 0	\$ 69,320	\$ 1,040	\$ -	\$ 17,280	\$ 40		\$ 69,320	\$ 325,281	21%
2024	Year 1	\$ 104,490	\$ 1,567	\$ -	\$ 34,130	\$ 79	97.5%	\$ 104,490	\$ 345,658	30%
2025	Year 2	\$ 141,382	\$ 2,121	\$ -	\$ 35,325	\$ 82	3.5%	\$ 141,382	\$ 366,748	39%
2026	Year 3	\$ 180,063	\$ 2,701	\$ 55,436	\$ 36,561	\$ 85	3.5%	\$ 180,063	\$ 388,576	46%
2027	Year 4	\$ 165,169	\$ 2,478	\$ 23,524	\$ 37,841	\$ 88	3.5%	\$ 165,169	\$ 360,637	46%
2028	Year 5	\$ 183,287	\$ 2,749	\$ -	\$ 39,165	\$ 91	3.5%	\$ 183,287	\$ 363,266	50%
2029	Year 6	\$ 226,572	\$ 3,399	\$ -	\$ 40,536	\$ 94	3.5%	\$ 226,572	\$ 387,467	58%
2030	Year 7	\$ 271,925	\$ 4,079	\$ -	\$ 41,954	\$ 97	3.5%	\$ 271,925	\$ 412,516	66%
2031	Year 8	\$ 319,427	\$ 4,791	\$ 368,838	\$ 43,423	\$ 101	3.5%	\$ 319,427	\$ 438,441	73%
2032	Year 9	\$ 323	\$ 5	\$ -	\$ 44,943	\$ 104	3.5%	\$ 323	\$ 176,781	0%
2033	Year 10	\$ 28,821	\$ 432	\$ 31,033	\$ 28,494	\$ 66	-36.6%	\$ 28,821	\$ 204,553	14%
2034	Year 11	\$ 27,712	\$ 416	\$ -	\$ 29,491	\$ 68	3.5%	\$ 27,712	\$ 209,774	13%
2035	Year 12	\$ 58,650	\$ 880	\$ -	\$ 30,523	\$ 71	3.5%	\$ 58,650	\$ 239,524	24%
2036	Year 13	\$ 91,122	\$ 1,367	\$ 124,999	\$ 31,591	\$ 73	3.5%	\$ 91,122	\$ 270,314	34%
2037	Year 14	\$ 186	\$ 3	\$ 8,417	\$ 32,697	\$ 76	3.5%	\$ 186	\$ 201,947	0%
2038	Year 15	\$ 25,613	\$ 384	\$ 18,429	\$ 33,842	\$ 78	3.5%	\$ 25,613	\$ 228,775	11%
2039	Year 16	\$ 42,595	\$ 639	\$ -	\$ 35,026	\$ 81	3.5%	\$ 42,595	\$ 250,732	17%
2040	Year 17	\$ 79,486	\$ 1,192	\$ -	\$ 36,252	\$ 84	3.5%	\$ 79,486	\$ 286,065	28%
2041	Year 18	\$ 118,199	\$ 1,773	\$ -	\$ 37,521	\$ 87	3.5%	\$ 118,199	\$ 322,635	37%
2042	Year 19	\$ 158,806	\$ 2,382	\$ 25,954	\$ 38,834	\$ 90	3.5%	\$ 158,806	\$ 360,485	44%
2043	Year 20	\$ 175,427	\$ 2,631	\$ 30,444	\$ 40,193	\$ 93	3.5%	\$ 175,427	\$ 379,731	46%
2044	Year 21	\$ 189,215	\$ 2,838	\$ -	\$ 41,600	\$ 96	3.5%	\$ 189,215	\$ 401,810	47%
2045	Year 22	\$ 235,109	\$ 3,527	\$ -	\$ 43,056	\$ 100	3.5%	\$ 235,109	\$ 443,775	53%
2046	Year 23	\$ 283,198	\$ 4,248	\$ 110,306	\$ 44,563	\$ 103	3.5%	\$ 283,198	\$ 487,208	58%
2047	Year 24	\$ 223,263	\$ 3,349	\$ 34,478	\$ 46,123	\$ 107	3.5%	\$ 223,263	\$ 440,425	51%
2048	Year 25	\$ 239,870	\$ 3,598	\$ 2,954	\$ 47,737	\$ 111	3.5%	\$ 239,870	\$ 466,387	51%
2049	Year 26	\$ 289,922	\$ 4,349	\$ -	\$ 49,408	\$ 114	3.5%	\$ 289,922	\$ 512,857	57%
2050	Year 27	\$ 345,408	\$ 5,181	\$ -	\$ 51,137	\$ 118	3.5%	\$ 345,408	\$ 562,698	61%
2051	Year 28	\$ 403,516	\$ 6,053	\$ 238,763	\$ 52,927	\$ 123	3.5%	\$ 403,516	\$ 614,283	66%
2052	Year 29	\$ 225,584	\$ 3,384	\$ 18,983	\$ 54,779	\$ 127	3.5%	\$ 225,584	\$ 523,094	43%
2053	Year 30	\$ 225,584	\$ 3,384	\$ 18,983	\$ 56,696	\$ 131	3.5%	\$ 266,681	\$ 565,838	47%



FULL FUNDING PLAN

STRUCTURAL INTEGRITY RESERVE STUDY

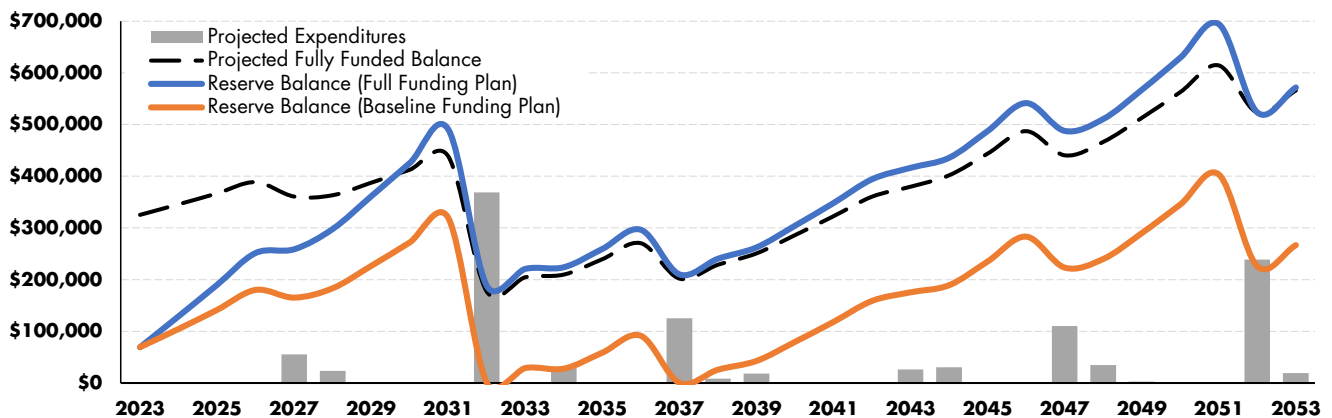
Year		Reserve Balance (Start of Year)	Investment Returns	Projected Expenditures	Reserve Contributions			Reserve Balance (End of Year)	Component Method	
					Association Annual Total	Average Unit Monthly Total	Change from Prior Year		Fully Funded Balance	Percent Funded
2023	Year 0	\$ 69,320	\$ 1,040	\$ -	\$ 17,280	\$ 40		\$ 69,320	\$ 325,281	21%
2024	Year 1	\$ 129,198	\$ 1,938	\$ -	\$ 58,838	\$ 136	240.5%	\$ 129,198	\$ 345,658	37%
2025	Year 2	\$ 189,975	\$ 2,850	\$ -	\$ 58,838	\$ 136	0.0%	\$ 189,975	\$ 366,748	52%
2026	Year 3	\$ 251,663	\$ 3,775	\$ 55,436	\$ 58,838	\$ 136	0.0%	\$ 251,663	\$ 388,576	65%
2027	Year 4	\$ 258,840	\$ 3,883	\$ 23,524	\$ 58,838	\$ 136	0.0%	\$ 258,840	\$ 360,637	72%
2028	Year 5	\$ 298,037	\$ 4,471	\$ -	\$ 58,838	\$ 136	0.0%	\$ 298,037	\$ 363,266	82%
2029	Year 6	\$ 361,346	\$ 5,420	\$ -	\$ 58,838	\$ 136	0.0%	\$ 361,346	\$ 387,467	93%
2030	Year 7	\$ 425,604	\$ 6,384	\$ -	\$ 58,838	\$ 136	0.0%	\$ 425,604	\$ 412,516	103%
2031	Year 8	\$ 490,827	\$ 7,362	\$ 368,838	\$ 58,838	\$ 136	0.0%	\$ 490,827	\$ 438,441	112%
2032	Year 9	\$ 188,189	\$ 2,823	\$ -	\$ 29,890	\$ 69	-49.2%	\$ 220,902	\$ 204,553	108%
2033	Year 10	\$ 220,902	\$ 3,314	\$ 31,033	\$ 30,936	\$ 72	3.5%	\$ 224,119	\$ 209,774	107%
2034	Year 11	\$ 224,119	\$ 3,362	\$ -	\$ 32,019	\$ 74	3.5%	\$ 259,499	\$ 239,524	108%
2035	Year 12	\$ 259,499	\$ 3,892	\$ -	\$ 33,139	\$ 77	3.5%	\$ 296,531	\$ 270,314	110%
2036	Year 13	\$ 296,531	\$ 4,448	\$ 124,999	\$ 34,299	\$ 79	3.5%	\$ 210,279	\$ 201,947	104%
2037	Year 14	\$ 210,279	\$ 3,154	\$ 8,417	\$ 35,500	\$ 82	3.5%	\$ 240,516	\$ 228,775	105%
2038	Year 15	\$ 240,516	\$ 3,608	\$ 18,429	\$ 36,742	\$ 85	3.5%	\$ 262,437	\$ 250,732	105%
2039	Year 16	\$ 262,437	\$ 3,937	\$ -	\$ 38,028	\$ 88	3.5%	\$ 304,402	\$ 286,065	106%
2040	Year 17	\$ 304,402	\$ 4,566	\$ -	\$ 39,359	\$ 91	3.5%	\$ 348,328	\$ 322,635	108%
2041	Year 18	\$ 348,328	\$ 5,225	\$ -	\$ 40,737	\$ 94	3.5%	\$ 394,289	\$ 360,485	109%
2042	Year 19	\$ 394,289	\$ 5,914	\$ 25,954	\$ 42,163	\$ 98	3.5%	\$ 416,413	\$ 379,731	110%
2043	Year 20	\$ 416,413	\$ 6,246	\$ 30,444	\$ 43,638	\$ 101	3.5%	\$ 435,853	\$ 401,810	108%
2044	Year 21	\$ 435,853	\$ 6,538	\$ -	\$ 45,166	\$ 105	3.5%	\$ 487,557	\$ 443,775	110%
2045	Year 22	\$ 487,557	\$ 7,313	\$ -	\$ 46,747	\$ 108	3.5%	\$ 541,617	\$ 487,208	111%
2046	Year 23	\$ 541,617	\$ 8,124	\$ 110,306	\$ 48,383	\$ 112	3.5%	\$ 487,818	\$ 440,425	111%
2047	Year 24	\$ 487,818	\$ 7,317	\$ 34,478	\$ 50,076	\$ 116	3.5%	\$ 510,733	\$ 466,387	110%
2048	Year 25	\$ 510,733	\$ 7,661	\$ 2,954	\$ 51,829	\$ 120	3.5%	\$ 567,268	\$ 512,857	111%
2049	Year 26	\$ 567,268	\$ 8,509	\$ -	\$ 53,643	\$ 124	3.5%	\$ 629,420	\$ 562,698	112%
2050	Year 27	\$ 629,420	\$ 9,441	\$ -	\$ 55,520	\$ 129	3.5%	\$ 694,382	\$ 614,283	113%
2051	Year 28	\$ 694,382	\$ 10,416	\$ 238,763	\$ 57,463	\$ 133	3.5%	\$ 523,498	\$ 523,094	100%
2052	Year 29	\$ 523,498	\$ 7,852	\$ 18,983	\$ 59,475	\$ 138	3.5%	\$ 571,842	\$ 565,838	101%
2053	Year 30									



COMPARISON OF CASH FLOW METHOD PLANS

STRUCTURAL INTEGRITY RESERVE STUDY

Year		Projected Expenditures	Baseline Funding			Full Funding		
			Annual Contribution	Reserve Balance (End of Year)	Percent Funded	Annual Contribution	Reserve Balance (End of Year)	Percent Funded
2023	Year 0		\$ 17,280	\$ 69,320	21%	\$ 17,280	\$ 69,320	21%
2024	Year 1	\$ -	\$ 34,130	\$ 104,490	30%	\$ 58,838	\$ 129,198	37%
2025	Year 2	\$ -	\$ 35,325	\$ 141,382	39%	\$ 58,838	\$ 189,975	52%
2026	Year 3	\$ -	\$ 36,561	\$ 180,063	46%	\$ 58,838	\$ 251,663	65%
2027	Year 4	\$ 55,436	\$ 37,841	\$ 165,169	46%	\$ 58,838	\$ 258,840	72%
2028	Year 5	\$ 23,524	\$ 39,165	\$ 183,287	50%	\$ 58,838	\$ 298,037	82%
2029	Year 6	\$ -	\$ 40,536	\$ 226,572	58%	\$ 58,838	\$ 361,346	93%
2030	Year 7	\$ -	\$ 41,954	\$ 271,925	66%	\$ 58,838	\$ 425,604	103%
2031	Year 8	\$ -	\$ 43,423	\$ 319,427	73%	\$ 58,838	\$ 490,827	112%
2032	Year 9	\$ 368,838	\$ 44,943	\$ 323	0%	\$ 58,838	\$ 188,189	106%
2033	Year 10	\$ -	\$ 28,494	\$ 28,821	14%	\$ 29,890	\$ 220,902	108%
2034	Year 11	\$ 31,033	\$ 29,491	\$ 27,712	13%	\$ 30,936	\$ 224,119	107%
2035	Year 12	\$ -	\$ 30,523	\$ 58,650	24%	\$ 32,019	\$ 259,499	108%
2036	Year 13	\$ -	\$ 31,591	\$ 91,122	34%	\$ 33,139	\$ 296,531	110%
2037	Year 14	\$ 124,999	\$ 32,697	\$ 186	0%	\$ 34,299	\$ 210,279	104%
2038	Year 15	\$ 8,417	\$ 33,842	\$ 25,613	11%	\$ 35,500	\$ 240,516	105%
2039	Year 16	\$ 18,429	\$ 35,026	\$ 42,595	17%	\$ 36,742	\$ 262,437	105%
2040	Year 17	\$ -	\$ 36,252	\$ 79,486	28%	\$ 38,028	\$ 304,402	106%
2041	Year 18	\$ -	\$ 37,521	\$ 118,199	37%	\$ 39,359	\$ 348,328	108%
2042	Year 19	\$ -	\$ 38,834	\$ 158,806	44%	\$ 40,737	\$ 394,289	109%
2043	Year 20	\$ 25,954	\$ 40,193	\$ 175,427	46%	\$ 42,163	\$ 416,413	110%
2044	Year 21	\$ 30,444	\$ 41,600	\$ 189,215	47%	\$ 43,638	\$ 435,853	108%
2045	Year 22	\$ -	\$ 43,056	\$ 235,109	53%	\$ 45,166	\$ 487,557	110%
2046	Year 23	\$ -	\$ 44,563	\$ 283,198	58%	\$ 46,747	\$ 541,617	111%
2047	Year 24	\$ 110,306	\$ 46,123	\$ 223,263	51%	\$ 48,383	\$ 487,818	111%
2048	Year 25	\$ 34,478	\$ 47,737	\$ 239,870	51%	\$ 50,076	\$ 510,733	110%
2049	Year 26	\$ 2,954	\$ 49,408	\$ 289,922	57%	\$ 51,829	\$ 567,268	111%
2050	Year 27	\$ -	\$ 51,137	\$ 345,408	61%	\$ 53,643	\$ 629,420	112%
2051	Year 28	\$ -	\$ 52,927	\$ 403,516	66%	\$ 55,520	\$ 694,382	113%
2052	Year 29	\$ 238,763	\$ 54,779	\$ 225,584	43%	\$ 57,463	\$ 523,498	100%
2053	Year 30	\$ 18,983	\$ 56,696	\$ 266,681	47%	\$ 59,475	\$ 571,842	101%



4.0 Discussion of Funding Plans

4.1 Funding Considerations

The decision on how an association chooses to fund its reserves is entirely up to the association and typically determined by an elected Board of Directors. In determining how to proceed, an association should consider many factors.

One important factor is the association's willingness to accept future special assessments. If an association's top priority is to avoid special assessments, then they should consider a conservative plan that is regularly updated, such as a Full Funding Plan with annual updates. In this scenario, it is nearly a guarantee that no special assessments will be required. Alternatively, if an association is agreeable to special assessments, they could consider a Baseline Funding Plan with updates only every ten years. We do not recommend this approach.

Another important element of reserve funding is maintaining an up-to-date reserve study. A detailed reserve study can often be updated for multiple years. A reserve study update is a simpler process than starting a full reserve study from scratch and can often be performed without the need for site visits and at a significantly lower cost.

4.2 Limitations of a Structural Integrity Reserve Study

As described in the Florida Statutes, a Structural Integrity Reserve Study (SIRS) is intended to provide an analysis of reserve funding for components related to the structural integrity and safety of buildings three stories and higher. Fundamentally, an SIRS only covers a portion of total association components. However, association components which are not related to the structural integrity and safety of the building can be numerous and costly to maintain, repair, and replace.

Association components outside of the scope of an SIRS can include:

- Site improvements such as roadways, sidewalks, asphalt, fencing, irrigation systems, etc.
- Components not part of a building, such as pools, amenity decks, etc.
- Structures less than three stories in height
- Elevator machinery and cabs
- Interiors amenities such as gyms, lobby finishes, corridor finishes, etc.
- Electronic amenities such as security systems, access control systems, etc.
- Mechanical systems such as HVAC components, cooling towers, vehicle gates, etc.

Therefore, if the Association chooses to follow one of the SIRS funding plans for its reserve contributions, we recommend that expenditures from that reserve account be limited to those included in the SIRS component list and for other components which are directly related to the structural integrity and safety of the building. In this scenario, a separate reserve account could be maintained and utilized for other association expenditures not related to the structural integrity and safety of the building.

It is also important to note that an SIRS is not intended to be a thorough structural evaluation or assessment but rather a useful tool to assist an association in its financial planning. The need for future structural repairs is not typically predictable and therefore such repairs may not meet the requirements to be considered reserve components. The need for structural repairs should be evaluated, at a minimum, during county-required building recertification process (if applicable) or state-required Milestone Inspection process. An SIRS is not the equivalent of a building recertification or other Milestone Inspection and will not meet the need of a building recertification and/or Milestone Inspection that may be required in the future.

5.0 Scope And Basis

5.1 Objectives

The purpose of a reserve study is to conduct a reserve analysis for the Association which includes an evaluation of the current rate of contribution to the reserves, and, if necessary, suggestions to alternative funding strategies. This report is intended to be used as a tool by the Association for considering and managing its future financial obligations, determining appropriate reserve fund allocations, and informing the individual owners of the Association's required reserve expenditures and the resulting financial plan.

For purposes of financial planning, Association expenses are typically divided into two categories:

- Annual Operations: Related to commonly held elements of real property and other assets. These expenses typically include annual operating expenditures, taxes, insurance, property management costs, and other service fees.
- Reserve Expenditures: Related to major periodic repairs and/or replacement of commonly held elements which meet the following criteria:
 - The Association has the obligation to maintain or replace the existing element.
 - The need and schedule for this project can be reasonably anticipated.
 - The total cost for the project is material to the Association, can be reasonably estimated, and includes all direct and related costs.

The focuses of our reserve analysis are long-term reserve expenditures, the funding plan, and ensuring adequate reserve balances. History demonstrates that, as time progresses, property conditions and management strategies will change. As a result, planned scopes of work may be altered or deferred, actual cost in the marketplace will vary from estimates, and actual rates of inflation and returns on investment will vary from projections. For these reasons, we concur with the CAI's guidelines and recommend that this reserve study be updated every three to five years.

The reserve expenditures included within the Structural Integrity Reserve Study differ from traditional reserve expenses and are described in the Florida Statutes. Please reference Section 5.2 of this report.

5.2 Level of Service

This analysis is consistent with the Florida Statutes section 718.112(2)(g)1, related to a Structural Integrity Reserve Study, which describes a reserve study encompassing “each building on the condominium property that is three stories or higher in height”, which will include the components within the SIRS Study Area and SIRS Item List “as related to the structural integrity and safety of the building”. Excerpts from the Florida Statutes are provided in Appendix B.

SIRS Study Area

The SIRS Study Area is limited to buildings three stories or higher and therefore excludes many common components which are not directly within (or affecting) buildings.

SIRS Item List

The SIRS Item List is consistent with the Florida Statutes section 718.112(2)(g)1 (a. through h.) which includes: the roof, structure, fireproofing and fire protection systems, plumbing, electrical systems, waterproofing and exterior painting, windows and exterior doors, and any other item that has a deferred maintenance expense or replacement cost that exceeds \$10,000 and the failure to replace or maintain such item negatively affects the items listed above. The SIRS Item List is not inclusive of all Association items within the building and excludes Association items such as interior finishes, interior amenities, HVAC systems, elevators, and other similar items not related to the structural integrity and safety of the building.

It is important to note that a Structural Integrity Reserve Study is not the equivalent of a building recertification or other Milestone Inspection and will not meet the need of a building recertification and/or Milestone Inspection that may be required in the future.

5.3 Opinions of Useful Life

For components which require periodic reserve expenditures for their repairs or replacement, the frequency of work equals the typical, industry accepted expected useful life (UL) for the type of feature. Theoretically, the remaining useful life (RUL) of a component, prior to the next reserve expenditure for its repair or replacement, is equal to the difference between its UL and its age: $RUL = UL - Age$.

In our experience, the effective age and actual RUL of an installed item can vary greatly from its actual age and calculated RUL. These variances depend on the quality of its original materials and workmanship, level of service, climatic exposure, and ongoing maintenance. As part of our analysis, we have determined our opinion of the UL and RUL of each common component based on our evaluation of its history and its existing condition.

In summary, we have based our opinion of the useful life, remaining useful life, and expected frequency and schedule of repair for each common component on some or all of the following:

- Actual or assumed age.
- Observed existing condition.
- Association's maintenance history and plan.
- Fannie Mae – Estimated Useful Life Tables
- Marshall & Swift Valuation Service Expected Life Expectancies
- Our experience with actual performance of such components under similar service and exposure.

5.4 Estimating

In developing our estimate of reserve expenditures for most common components, we have estimated quantities of each item as well as unit costs for its repair or replacement. In some cases, it is more appropriate to estimate a lump sum cost for a required work package or 'lot'. In other cases, particularly when future expenditures are difficult to predict, it is more appropriate to allocate an allowance.

Unless directed to take a different approach, we assume that contract labor will perform the work and apply appropriate installer's mark-ups on supplied material and equipment. When required, our estimated costs consider demolition and disposal of existing materials as well as protection of other portions of the property. When appropriate for large reserve projects, we will also consider soft costs for design and project management, as well as typical general contractor's cost for general conditions, supervision, overhead and profit.

We have based our opinion of unit, lump sum, and allowance costs on some or all of the following:

- Records of previous expenses.
- Previously solicited Vendor quotations or Contractor proposals.
- Provided reserve budgets developed by others.
- Our project files on repairs and replacements at other properties.
- RSMeans Construction Cost Data.
- Marshall & Swift Valuation Service – Facility Cost Index.

The primary purpose of this reserve study is to aid in overall budget planning. While we make every effort to be accurate, our cost and quantity estimates are to be considered preliminary and by no means a guarantee. Annual reserve expenditure budgets have been calculated for all years during the study period by inflating the annual tallies of current dollar cost at a rate of inflation selected by the Association.

5.5 Information Provided

Our initial financial analysis was based on the current funding rate and future plans of the Association. We were provided with initial information on the reserves and its funding plan.

- Reserve Balance Provided: \$65,000
- As of: September 29, 2023
- Current Monthly Reserve Contribution: \$1,440
- Fiscal Year Starting Date: January 1, 2024
- For Designated Year: 2024
- Projected Reserve Balance on Starting Date: \$69,320
- Planned Increases: N/A
- Planned Special Assessments: N/A
- Projected Annual Return on Investment: 1.5%
- Projected Rate of Annual Inflation: 3.5%

Financial data, records of past expenses, and cost estimates provided by others have been taken in good faith and at face value. No audit or other verification has been performed.

6.0 Annual Projected Expenditures

PROJECTED EXPENDITURE SCHEDULE (PAGE 1 OF 3)

#	Component Description	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
		Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Building Structure and Envelope											
101	Roofing and appurtenances, replace	-	-	-	-	-	-	-	-	\$ 360,279	-
102	Roofing, silicone coating	-	-	-	-	-	-	-	-	-	-
103	Waterproof coating, walkways	-	-	-	-	-	-	-	-	-	-
104	Exterior painting, waterproofing, and minor repairs	-	-	-	\$ 55,436	-	-	-	-	-	-
105	Exterior windows, storage rooms	-	-	-	-	-	-	-	-	-	-
106	Exterior metal doors, mechanical rooms	-	-	-	-	-	-	-	-	-	-
107	Structural repairs, milestone inspection	-	-	-	-	-	-	-	-	-	-
Electrical Systems											
201	Main circuit breaker, 1200 amps, 120/208V, 3P, 4W	-	-	-	-	-	-	-	-	-	-
202	Electrical subpanels, 60-225 amps	-	-	-	-	-	-	-	-	-	-
203	Electrical disconnects, electrical room, 60-225 amps	-	-	-	-	-	-	-	-	-	-
204	Electrical disconnects, elevator room, 60-225 amps	-	-	-	-	-	-	-	-	-	-
Plumbing											
301	Commercial water heater, 18 kW	-	-	-	-	-	-	-	-	-	-
302	Vertical storage tank, 119 gallons	-	-	-	-	-	-	-	-	-	-
303	Domestic water, backflow preventer	-	-	-	-	-	-	-	-	-	-
304	Domestic water, backflow preventer	-	-	-	-	-	-	-	-	\$ 8,559	-
Fire Protection Systems											
401	Fire alarm panel	-	-	-	-	\$ 15,492	-	-	-	-	-
402	Fire alarm system components	-	-	-	-	\$ 8,033	-	-	-	-	-
403	Fire department connections	-	-	-	-	-	-	-	-	-	-
Grand Total		-	-	-	\$ 55,436	\$ 23,524	-	-	-	\$ 368,838	-

PROJECTED EXPENDITURE SCHEDULE (PAGE 2 OF 3)

#	Component Description	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043
		Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20
Building Structure and Envelope											
101	Roofing and appurtenances, replace	-	-	-	-	-	-	-	-	-	-
102	Roofing, silicone coating	-	-	-		-	-	-	-	-	-
103	Waterproof coating, walkways				\$ 18,650		-	-	-	-	-
104	Exterior painting, waterproofing, and minor repairs	-			\$ 78,198		-	-	-	-	-
105	Exterior windows, storage rooms	-		-		-		-	-	-	-
106	Exterior metal doors, mechanical rooms					-	-	-	-	-	-
107	Structural repairs, milestone inspection	-		-	\$ 28,151	-	-		-	-	-
Electrical Systems											
201	Main circuit breaker, 1200 amps, 120/208V, 3P, 4W	-	-	-		-		-	-	-	-
202	Electrical subpanels, 60-225 amps	-	-	-			\$ 10,052		-	-	-
203	Electrical disconnects, electrical room, 60-225 amps					-	\$ 8,377		-	-	-
204	Electrical disconnects, elevator room, 60-225 amps	-	-	-	-	-		-	-	-	-
Plumbing											
301	Commercial water heater, 18 kW	\$ 14,811	-	-	-	-	-	-	-	-	-
302	Vertical storage tank, 119 gallons	-	-	-		\$ 8,417	-	-	-	-	-
303	Domestic water, backflow preventer	\$ 4,937				-	-	-	-	-	-
304	Domestic water, backflow preventer	-		-		-	-	-	-	-	-
Fire Protection Systems											
401	Fire alarm panel	-	-	-	-	-	-	-	-		\$ 25,954
402	Fire alarm system components				-		-	-	-	-	-
403	Fire department connections	\$ 11,285		-		-	-	-	-	-	-
Grand Total		\$ 31,033	-	-	\$ 124,999	\$ 8,417	\$ 18,429	-	-	-	\$ 25,954

PROJECTED EXPENDITURE SCHEDULE (PAGE 3 OF 3)

#	Component Description	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053
		Year 21	Year 22	Year 23	Year 24	Year 25	Year 26	Year 27	Year 28	Year 29	Year 30
Building Structure and Envelope											
101	Roofing and appurtenances, replace	-	-	-	-	-	-	-	-	-	-
102	Roofing, silicone coating	-	-	-	-	-	-	-	-	\$ 207,518	-
103	Waterproof coating, walkways	-	-	-	-	-	-	-	-	\$ 31,246	-
104	Exterior painting, waterproofing, and minor repairs	-	-	-	\$ 110,306	-	-	-	-	-	-
105	Exterior windows, storage rooms	-	-	-	-	\$ 34,478	-	-	-	-	-
106	Exterior metal doors, mechanical rooms	\$ 7,561	-	-	-	-	-	-	-	-	-
107	Structural repairs, milestone inspection	-	-	-	-	-	-	-	-	-	-
Electrical Systems											
201	Main circuit breaker, 1200 amps, 120/208V, 3P, 4W	\$ 22,883	-	-	-	-	-	-	-	-	-
202	Electrical subpanels, 60-225 amps	-	-	-	-	-	-	-	-	-	-
203	Electrical disconnects, electrical room, 60-225 amps	-	-	-	-	-	-	-	-	-	-
204	Electrical disconnects, elevator room, 60-225 amps	-	-	-	-	-	\$ 2,954	-	-	-	-
Plumbing											
301	Commercial water heater, 18 kW	-	-	-	-	-	-	-	-	-	-
302	Vertical storage tank, 119 gallons	-	-	-	-	-	-	-	-	-	-
303	Domestic water, backflow preventer	-	-	-	-	-	-	-	-	-	-
304	Domestic water, backflow preventer	-	-	-	-	-	-	-	-	-	-
Fire Protection Systems											
401	Fire alarm panel	-	-	-	-	-	-	-	-	-	-
402	Fire alarm system components	-	-	-	-	-	-	-	-	-	\$ 18,983
403	Fire department connections	-	-	-	-	-	-	-	-	-	-
Grand Total		\$ 30,444	-	-	\$ 110,306	\$ 34,478	\$ 2,954	-	-	\$ 238,763	\$ 18,983

7.0 Standards and Limitations

Criterium-Cromer Engineers shall perform duties to at least the professional standards consistent with a licensed, Professional Engineer, but does not guarantee or warrant that all adverse conditions concerning the property can be or will be discovered and included in the report. The photographs are an integral part of this report and must be included in any review.

This study is limited to the visual observations made during our inspection. We did not undertake any excavation, conduct any destructive or invasive testing, remove surface materials or finishes, or displace furnishings or equipment. The observations described in this study are valid on the dates of the investigation. Accordingly, we cannot comment on the condition of systems that we could not see, such as buried structures and utilities, nor are we responsible for conditions that could not be seen or were not within the scope of our services at the time of inspection. We did not perform any computations or other engineering analysis as part of this study, nor did we conduct a comprehensive code compliance investigation.

This information in this study is not to be considered a warranty of condition, quality, compliance or cost. No warranty is implied. Financial data, records of past expenses, and cost estimates provided by others have been taken in good faith and at face value. No audit or other verification has been performed.

Reserve budgets are opinions of likely expense based on reasonable cost estimates. We have not obtained competitive quotations or estimates from contractors. Actual costs can vary significantly, based on the specific scope of work developed, availability of materials and qualified contractors, and many other variables. We cannot be responsible for variances.

Criterium-Cromer Engineers does not offer financial counseling services. Although reasonable rates of inflation and return on investment must be assumed to calculate projected balances, no one can accurately predict actual economic performance. Although reserve fund management and investment may be discussed during the course of the study, we do not purport to hold any special qualifications in this area. We recommend that the Association also seek other professional guidance before finalizing their current reserve fund planning activity. Depending on issues which may arise, an appropriate team of consultants to aid decision-making might include their property manager, accountant, financial counselor, and attorney.

Criterium-Cromer Engineers prepared this confidential report for the review and use of the Association. We do not intend any other individual or party to rely upon this study without our express written consent. If another individual or party relies on this study, they shall indemnify, defend and hold Criterium-Cromer Engineers, its subsidiaries, affiliates, officers, directors, members, shareholders, partners, agents, employees and such other parties in interest specified by Criterium-Cromer Engineers harmless for any damages, losses, or expenses they may incur as a result of its use. Any use or reliance of the report by an individual or party other than shall constitute acceptance of these terms and conditions.

8.0 Conclusion

To the best of our ability, we have attempted to work in the best interest of the Association and to aid the Board toward fulfillment of their fiduciary responsibilities and obligations to the individual Unit Owners who comprise the association's membership. In our professional opinion, and within the limitations disclosed elsewhere herein, all information contained herein is reliable and appropriate to guide the Board's deliberations and decision-making. We consider our report confidential and will not share its content with anyone but the Board without its knowledge and release.

We are unaware of any other involvement or business relationship between Criterium-Cromer Engineers and the Developer, individual Unit Owners, members of the Board, or any other entities which constitutes any conflict of interest.

Criterium Engineers appreciates this opportunity to assist you in support of the financial planning for your community. We are pleased to present this report for the Board's consideration and use. Thank you.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "Casey Cromer".

Casey Cromer, P.E.
President
Criterium-Cromer Engineers

APPENDIX A

COMPONENT PHOTOGRAPHS

APPENDIX B

REFERENCE DOCUMENTS

EXCERPTS FROM THE FLORIDA STATUTES

Section 718.112(2)(g): Structural Integrity Reserve Study –

1. A residential condominium association must have a structural integrity reserve study completed at least every 10 years after the condominium's creation for each building on the condominium property that is three stories or higher in height, as determined by the Florida Building Code, which includes, at a minimum, a study of the following items as related to the structural integrity and safety of the building:
 - a. Roof.
 - b. Structure, including load-bearing walls and other primary structural members and primary structural systems as those terms are defined in s. 627.706.
 - c. Fireproofing and fire protection systems.
 - d. Plumbing.
 - e. Electrical systems.
 - f. Waterproofing and exterior painting.
 - g. Windows and exterior doors.
 - h. Any other item that has a deferred maintenance expense or replacement cost that exceeds \$10,000 and the failure to replace or maintain such item negatively affects the items listed in sub-subparagraphs a.-g., as determined by the visual inspection portion of the structural integrity reserve study.
2. A structural integrity reserve study is based on a visual inspection of the condominium property. A structural integrity reserve study may be performed by any person qualified to perform such study. However, the visual inspection portion of the structural integrity reserve study must be performed or verified by an engineer licensed under chapter 471, an architect licensed under chapter 481, or a person certified as a reserve specialist or professional reserve analyst by the Community Associations Institute or the Association of Professional Reserve Analysts.
3. At a minimum, a structural integrity reserve study must identify each item of the condominium property being visually inspected, state the estimated remaining useful life and the estimated replacement cost or deferred maintenance expense of each item of the condominium property being visually inspected, and provide a reserve funding schedule with a recommended annual reserve amount that achieves the estimated replacement cost or deferred maintenance expense of each item of condominium property being visually inspected by the end of the estimated remaining useful life of the item. The structural integrity reserve study may recommend that reserves do not need to be maintained for any item for which an estimate of useful life and an estimate of replacement cost cannot be determined, or the study may recommend a deferred maintenance expense amount for such item. The structural integrity reserve study may recommend that reserves for replacement costs do not need to be maintained for any item with an estimated remaining useful life of greater than 25 years, but the study may recommend a deferred maintenance expense amount for such item.
4. This paragraph does not apply to buildings less than three stories in height; single-family, two-family, or three-family dwellings with three or fewer habitable stories above ground; any portion or component of a building that has not been submitted to the condominium form of ownership; or any portion or component of a building that is maintained by a party other than the association.

Reserve Study Standards®™

RSS - RS052023

Terms and Definitions

Adequate Reserves: A replacement reserve fund and stable and equitable multiyear [funding plan](#) that together provide for the reliable and timely execution of the association's major repair and replacement projects as defined herein without reliance on additional supplemental funding.

Capital Improvements: Additions to the association's common area that previously did not exist. While these components should be added to the reserve study for future replacement, the cost of construction or installation cannot be taken from the reserve fund.

Cash Flow Method (also known as pooling): A method of developing a reserve funding plan where funding of reserves is designed to offset the annual expenditures from the reserve fund.

To determine the selected funding plan, different reserve funding plans are tested against the anticipated schedule of reserve expenses until the desired funding goal is achieved.

Common Area: The areas identified in the community association's master deed or declarations of covenant easements and restrictions that the association is obligated to maintain and replace or based on a well-established association precedent.

Community Association: A nonprofit entity that exists to preserve the nature of the community and protect the value of the property owned by members. Membership in the community association is mandatory and automatic for all owners. All owners pay mandatory lien-based assessments that fund the operation of the association and maintain the common area or elements, as defined in the governing documents. The community association is served and lead by an elected board of trustees or directors.

Components: The individually listed projects within the physical analysis which are determined for inclusion using the process described within the component inventory. These components form the building blocks for the reserve study. **Components are selected to be included in the reserve study based on the following three-part test:**

1. The association has the obligation to maintain or replace the existing element.
2. The need and schedule for this project can be reasonably anticipated.
3. The total cost for the project is material to the association, can be reasonably estimated, and includes all direct and related costs.

Component Inventory: The task of selecting and quantifying reserve components. This task can be accomplished through on-site visual observations, review of association design and organizational documents, review of association precedents, and discussion with appropriate representative(s) of the association.

The Reserve Specialist, in coordination with the client, will determine the methodology for including these components in the study. Typical evaluation techniques for consideration include:

- Inclusion of long-life components with funding in the study.
- Addition of long-life components with funding at the time when they fall within the 30-year period from the date of study preparation.

- Identification of long-life components in the component inventory even when they are not yet being funded in the 30-year funding plan.

Component Method (also known as Straight Line): A method of developing a reserve funding plan where the total funding is based on the sum of funding for the individual components.

Condition Assessment: The task of evaluating the current condition of the component based on observed or reported characteristics. The assessment is limited to a visual, non-invasive evaluation.

Effective Age: The difference between [useful life](#) and estimated [remaining useful life](#). Not always equivalent to chronological age since some components age irregularly. Used primarily in computations.

Financial Analysis: The portion of a reserve study in which the current status of the reserves (measured as cash or [percent funded](#)) and a recommended reserve funding plan are derived, and the projected reserve income and expense over a period of time are presented. The financial analysis is one of the two parts of a reserve study. A minimum of 30 years of income and expense are to be considered.

Fully Funded: 100 percent funded. When the actual (or projected) [reserve balance](#) is equal to the fully funded balance.

Fully Funded Balance (FFB): An indicator against which the actual (or projected) reserve balance can be compared. The reserve balance that is in direct proportion to the fraction of life “used up” of the current repair or [replacement cost](#). This number is calculated for each component, and then summed for an association total.

$$\text{FFB} = \text{Current Cost} \times \text{Effective Age} / \text{Useful Life}$$

Example: For a component with a \$10,000 current replacement cost, a 10-year useful life, and effective age of 4 years, the fully funded balance would be \$4,000.

Fund Status: The status of the reserve fund reported in terms of cash or [percent funded](#).

Funding Goals:

The three funding goals listed below range from the most aggressive to most conservative:

Baseline Funding

Establishing a reserve funding goal of allowing the reserve cash balance to approach but never fall below zero during the cash flow projection. This is the funding goal with the greatest risk of being prepared to fund future repair and replacement of major components, **and it is not recommended** as a long-term solution/plan. Baseline funding may lead to project delays, the need for a [special assessment](#), and/or a line of credit for the community to fund needed repairs and replacement of major components.

Threshold Funding

Establishing a reserve funding goal of keeping the [reserve balance](#) above a specified dollar or percent funded amount. Depending on the threshold selected, this funding goal may be weaker or stronger than “fully funded” with respective higher risk or less risk of cash problems. In determining the threshold, many variables should be considered, including things such as

investment risk tolerance, community age, building type, components that are not readily inspected, and components with a [remaining useful life](#) of more than 30 years.

Full Funding

Setting a reserve funding goal to attain and maintain reserves at or near 100 percent funded. Fully funded is when the actual or projected reserve balance is equal to the fully funded balance.

It should be noted that, in certain jurisdictions, there may be statutory funding requirements that would dictate the funding requirements. In all cases, these standards are considered the minimum to be referenced.

Funding Plan: An association's plan to provide income to a reserve fund to offset anticipated expenditures from that fund. The plan must be a minimum of 30 years of projected income and expenses.

Funding Principles: A funding plan addressing these principles. These funding principles are the basis for the recommendations included within the reserve study:

- Sufficient funds when required.
- Stable funding rate over the years.
- Equitable funding rate over the years.
- Fiscally responsible.

Initial Year: The first fiscal year in the financial analysis or funding plan.

Life Estimates: The task of estimating [useful life](#) and [remaining useful life](#) of the reserve components.

Life Cycle Cost: The ongoing cost of deterioration which must be offset in order to maintain and replace common area components at the end of their useful life. Note that the cost of preventive maintenance and corrective maintenance determined through periodic structural inspections (if required) are included in the calculation of life cycle costs and often result in overall net lower life cycle costs.

Maintenance: Maintenance is the process of maintaining or preserving something, or the state of being maintained. Maintenance is often defined in three ways: preventive maintenance, corrective maintenance, and deferred maintenance. Maintenance projects commonly fall short of "replacement" but may pass the defining test of a reserve component and be appropriate for reserve funding. Maintenance types are categorized below:

Preventive Maintenance: Planned maintenance carried out proactively at predetermined intervals, aimed at reducing the performance degradation of the component such that it can attain, at minimum, its estimated useful life.

Deferred Maintenance: Maintenance which is not performed and leads to premature deterioration to the common areas due to lack of preventive maintenance.

This results in a reduction in the remaining useful life of the reserve components and the potential of inadequate funding. Typically, deferred maintenance creates a need for corrective maintenance.

Corrective Maintenance: Maintenance performed following the detection of a problem, with the goal of remediating the condition such that the intended function and life of the component or system is restored, preserved, or enhanced.

Many corrective maintenance projects could be prevented with a proactive, preventive maintenance program. Note that when the scope is minor, these projects may fall below the threshold of cost significance and thus are handled through the operational budget. In other cases, the cost and timing should be included within the reserve study.

Percent Funded: The ratio, at a particular point in time clearly identified as either the beginning or end of the association's fiscal year, of the actual (or projected) [reserve balance](#) to the fully funded balance, expressed as a percentage.

While percent funded is an indicator of an association's reserve fund size, it should be viewed in the context of how it is changing due to the association's reserve funding plan, in light of the association's risk tolerance and is not by itself a measure of "adequacy."

Periodic Structural Inspection: [Structural system](#) inspections aimed at identifying issues when they become evident.

Additional information and recommendations are included within the Condominium Safety Public Policy Report. www.condosafety.com

Physical Evaluation: The portion of the reserve study where the component inventory, condition assessment, and life and [valuation estimate](#) tasks are performed. This represents one of the two parts of the reserve study.

Preventive Maintenance Schedule: A summary of the preventive maintenance tasks included within a maintenance manual which should be performed such that the useful lives of the components are attained or exceeded. This schedule should include both the timing and the estimated cost of the task(s).

Remaining Useful Life (RUL): Also referred to as "remaining life" (RL). The estimated time, in years, that a component can be expected to serve its intended function, presuming timely preventive maintenance. Projects expected to occur in the initial year have zero remaining useful life.

Replacement Cost: The cost to replace, repair, or restore the component to its original functional condition during that particular year, including all related expenses (including but not limited to shipping, engineering, design, permits, installation, disposal, etc.).

Reserve Balance: Actual or projected funds, clearly identified as existing either at the beginning or end of the association's fiscal year, which will be used to fund reserve component expenditures. The source of this information should be disclosed within the reserve study.

Also known as beginning balance, reserves, reserve accounts, or cash reserves. This balance is based on information provided and not audited.

Reserve Study: A reserve study is a budget planning tool which identifies the components that a community association is responsible to maintain or replace, the current status of the reserve fund, and a stable and equitable funding plan to offset the anticipated future major common area expenditures.

This limited evaluation is conducted for budget and cash flow purposes. Tasks outside the scope of a reserve study include, but are not limited to, design review, construction evaluation, intrusive or destructive testing, preventive maintenance plans, and structural or safety evaluations.

Reserve Study Provider: An individual who prepares reserve studies. In many instances, the reserve study provider will possess a specialized designation such as the Reserve Specialist® (RS) designation administered by Community Associations Institute (CAI). This designation indicates that the provider has shown the necessary skills to perform a reserve study that conforms to these standards. In some instances, qualifications in excess of the RS designation will be required if supplemental subject matter expertise is required.

Reserve Study Provider Firm: A company that prepares reserve studies as one of its primary business activities.

Responsible Charge: A Reserve Specialist (RS) in responsible charge of a reserve study shall render regular and effective supervision to those individuals' performing services that directly and materially affect the quality and competence of services rendered by the Reserve Specialist. A Reserve Specialist shall maintain such records as are reasonably necessary to establish that the Reserve Specialist exercised regular and effective supervision of a reserve study of which he or she was in responsible charge. A Reserve Specialist engaged in any of the following acts or practices shall be deemed not to have rendered the regular and effective supervision required herein:

1. The regular and continuous absence from principal office premises from which professional services are rendered; except for performance of field work or presence in a field office maintained exclusively for a specific project;
2. The failure to personally inspect or review the work of subordinates where necessary and appropriate;
3. The rendering of a limited, cursory or perfunctory review of plans or projects in lieu of an appropriate detailed review; and
4. The failure to personally be available on a reasonable basis or with adequate advance notice for consultation and inspection where circumstances require personal availability.

Site Visit: A visual assessment of the accessible areas of the components included within the reserve study.

The site visit includes tasks such as, but not limited to, on-site visual observations, a review of the association's design and governing documents, review of association precedents, and discussion with appropriate representative(s) of the association.

Special Assessment: A temporary assessment levied on the members of an association in addition to regular assessments. Note that special assessments are often regulated by governing documents or local statutes.

Special assessments, when used to make up for unplanned reserve fund shortfalls, may be an indicator of deferred maintenance, improper reserve project planning, and unforeseen catastrophes and accidents, as well as other surprises.

Structural System: The structural components within a building that, by contiguous interconnection, form a path by which external and internal forces, applied to the building, are delivered to the ground. This is generally a combination of structural beams, columns, and bracing and is not included within the reserve study, although it is reviewed as part of the recommended periodic structural inspections.

It is important to recognize that individual structural components which are not a part of the structural system, such as decks, balconies, and podium deck components may be included for reserve funding if they otherwise satisfy the three-part test.

Useful Life (UL): The estimated time, in years, that a reserve component can be expected to serve its intended function if properly constructed presuming proactive, planned, preventive maintenance.

Best practice is that a component's Useful Life should reflect the actual preventive maintenance being performed (or not performed).

Valuation Estimates: The task of estimating the current repair or [replacement costs](#) for the reserve components.

APPENDIX C

QUALIFICATIONS



BUILDING INSPECTION ENGINEERS
PROUDLY SERVING NORTH AMERICA SINCE 1957

Casey Cromer, P.E.
President / Senior Engineer



Casey Cromer is a Professional Engineer and a native of Miami, Florida. Within the Greater Miami area, he has over 10 years of forensic engineering and construction experience and has consulted on projects varying from single-family residences, to high-rise condominiums, to construction collapses. His primary focus has been the building envelope including stucco, fenestrations, roofing, waterproofing, concrete protection, and more.

His practice areas include forensic investigations, expert witness services, litigation support, fenestrations, waterproofing, claddings, roofing, property condition assessments, post-tension cable inspections, aerial drone surveys, and more.

EDUCATION, PROFESSIONAL LICENSES, AND AFFILIATIONS

The University of Central Florida (UCF) – Orlando, Florida

Bachelor of Science, Civil Engineering

Licensed Professional Engineer

State of Florida, No. 87329

ASTM Committee C11 (Gypsum and Related Building Materials and Systems), Voting Member

Stucco Institute – Sealed Stucco System Technician

FAA Certified Remote Pilot (Drone)

WHY I DO WHAT I DO

"I have always been fascinated by buildings and have a knack for problem solving – so what better way to put those two together than engineering! Being able to consult and work on buildings of all sizes has kept my days interesting and the knowledge gained throughout each project has been invaluable."

PROJECT HIGHLIGHTS

Building Envelope Investigations

- Acqualina Resort & Residences (52-story, five-star hotel and residence) – Entire façade (stucco and EIFS) via swing stage and drone for structural integrity of wall claddings, including recommendations and repairs.
- Akoya Condominium (47-story luxury condo) – Entire façade (stucco) via swing stage for structural wall repairs during stucco remediation. Daily inspections to sign off on structural repairs.
- 800 Waterford (250,000 sq. ft. Class-A office) – Entire façade (precast panels and curtainwalls) via swing stage for water tightness including water testing of repaired areas.

Construction Defect Projects

- Represented developers, general contractors, subcontractors, associations, and/or owners.
- Acqualina Mansions, Aria (Longboat Key, FL), Aria on the Bay, Asia Brickell Key, Axis Brickell, Chateau Beach, CitySide (Sarasota, FL), Continuum South Beach, Crimson Condo, Eloquence on the Bay, Fendi Chateau, Jade Ocean, Grove at Grand Bay, Iconbay Condo, Mint Condo, Nine at Mary Brickell, Nordica Condo, Ocean 7, Peninsula Aventura, Toscano at Dadeland, Vista Del Mar (Myrtle Beach, SC), Vizcayne, W South Beach, 1100 Millecento, 321 Ocean, 900 Biscayne...

Structural Collapse Investigations

- Champlain Towers (Surfside, FL) – Investigated and lead team of subject experts for in-service condominium collapse.
- Miami-Dade College Parking Garage – construction collapse investigation to determine collapse origin, original design requirements, and repair feasibility.
- Key West International Airport – Litigation support for a construction collapse.

Property Loss Investigations

- Cause and origin investigations on hundreds of residential and commercial buildings.
- Determine structural and material failures as related to construction defects, storm damage, water intrusion, normal wear and tear, etc.



BUILDING INSPECTION ENGINEERS
PROUDLY SERVING NORTH AMERICA SINCE 1957

H. Alan Mooney, P.E. *Founding President*



Alan Mooney is a civil and structural engineer with over 40 years of experience as a consulting engineer. From 1988 until 2018 he was President and principal owner of Criterium Engineers, a national consulting engineering firm with affiliate offices throughout North America.

His experience includes:

- complex multi-million-dollar engineering and construction projects
- forensic engineering
- construction quality assurance services
- numerous building envelope quality assurance and commissioning projects
- thousands of residential and commercial building inspections

In addition to his own projects, he continues to serve as an advisor/consultant for inspections, structural evaluations, investigative engineering, structural design and trainer for the Criterium Engineers staff.

As a structural engineer, he has designed a variety of structures in wood, concrete and steel. These structures include bridges, multi-story buildings, parking garages and marine facilities.

Mr. Mooney has also established an impressive track record as a noted seminar leader and author, both locally and nationally, on construction-related issues, construction quality, and building inspection procedures and standards.

EDUCATION AND PROFESSIONAL AFFILIATION

Rutgers University, New Brunswick, NJ – 1969
Bachelors of Science, Civil Engineering

Licensed Professional Engineer in ME, NH, VT, MA, CT, NY, NC, NJ, AZ, NV, FL, KS, WA
CAI Reserve Specialist (RS)
Licensed Reserve Study Specialist in NV

CAI (Community Associations Institute)
ASCE (American Society of Civil Engineers)
The Order of the Engineer

WHY I DO WHAT I DO

"Building technology is always changing; keeping up is an exciting challenge. Diagnosing problems means using good judgment and capitalizing on years of experience. It's even more challenging and exciting because every client's needs are different. What we do represents the essence of being a professional engineer."

WHY CRITERIUM ENGINEERS

"I founded Criterium Engineers to allow other engineers to discover their full potential as professionals."

SELECT PROJECT HIGHLIGHTS

- **Silo Point, Baltimore, Maryland** – provided transition study and follow-up consulting for a unique, high-end condominium complex involving the conversion of an abandoned grain handling complex.
- **Sun City Anthem, Las Vegas, Nevada** – provided comprehensive reserve fund study for a large (10,000 residents), high end home owner association in Las Vegas.
- **San Diego Airport Expansion** – envelope commissioning
- **Phoenix Sky Harbor Airport** – envelope quality assurance
- **IKEA** – facilities review of all locations in the U.S.
- **Wimar-Tahoe** – provided expert testimony for building performance in a \$100 million dispute involving a Lake Tahoe casino complex.
- **American Residential Properties** – provided property evaluation reports for a national client purchasing thousands of homes as rental properties across the country

EXPERIENCE HIGHLIGHTS

- 30 years' experience as a construction quality consultant including collaboration with several major builders to develop effective quality assurance programs.
- 30 years' experience as a construction expert in construction disputes, including serving as an expert witness on numerous occasions.
- Has performed more than 15,000 building inspections personally
- Criterium Engineers now performs over 20,000 building inspections annually to standards Mr. Mooney developed and refines on an ongoing basis
- Founding president of the National Academy of Building Inspection Engineers (NABIE), 1989-1993.
- Co-author of the NABIE Standards of Practice for Home Inspections
- Author of the National Association of Home Builders (NAHB) Quality Construction for the Master Builder
- 30 years experience as a seminar leader; presented seminars to builders, appraisers, real estate agents in more than 30 states
- National and regional speaker for CAI conferences
- Chosen Speaker of the Year by the Dallas, TX chapter of the Community Associations Institute (CAI), 2001